

Homework 1: Linear Consensus

Due date: September 24, 2025, 11:59 PM on Gradescope

Problem 1: Graphs and Linear Consensus Dynamics (50 pts)

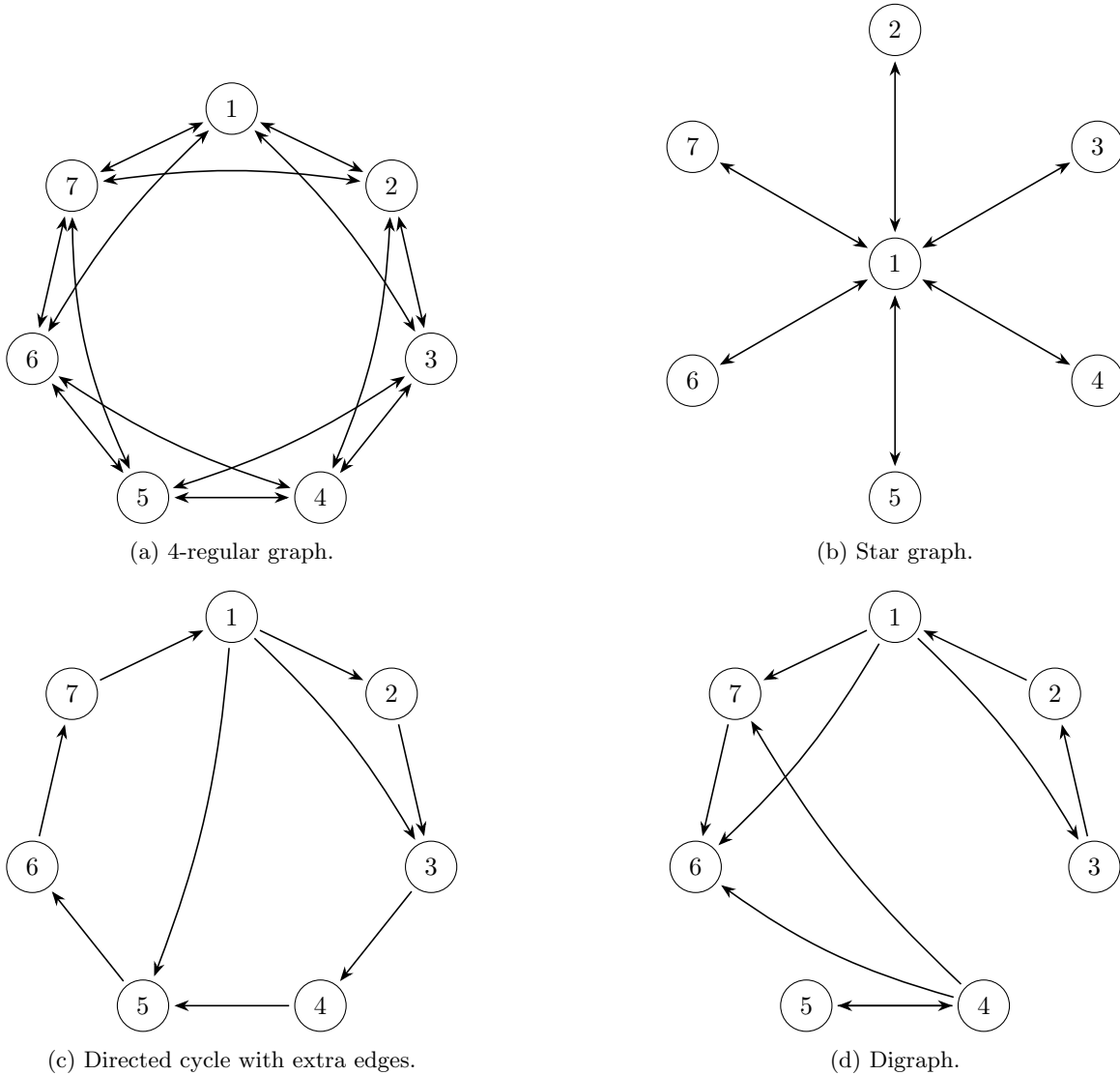


Figure 1: Four 7-node graphs with directed edges; arrow direction indicates direction of information flow

1. (15 pts) For each of the four graphs shown in Figure 1, answer the following:

- (a) Assuming each graph is unweighted, i.e. $a_{ij} \in \{0, 1\}$ where $a_{ij} = 1$ iff there is an edge $j \rightarrow i$, write down the corresponding adjacency matrix A , in-degree matrix Δ , and the graph Laplacian matrix L . Is the graph strongly connected?

- (b) Compute all of the eigenvalues of the Laplacian matrix.
 - (c) Find $\mathbf{w}$, a left null eigenvector of L satisfying $\mathbf{w}^T L = 0$, $\mathbf{w}^T \mathbf{1} = 1$ where $\mathbf{1} = (1, 1, \dots, 1)^T$.
 - (d) Is the graph weight-balanced? If not, come up with an alternative set of positive edge weights $\hat{a}_{ij} > 0$ which make the pictured digraph weight-balanced. Write down the resulting weighted adjacency matrix $\hat{A}$, in-degree matrix $\hat{\Delta}$, and Laplacian matrix $\hat{L}$. Alternatively, explain why it is not possible to do so.
2. (10 pts) We say a network converges to a *consensus* when its state $\mathbf{x}(t) \rightarrow \alpha \mathbf{1}$ as $t \rightarrow \infty$.
- (a) Based on your calculations in Part 1 (a)-(c), which of the pictured unweighted graphs do you expect to reach consensus under a linear protocol? Explain.
 - (b) For an initial condition $\mathbf{x}(0) = (1, 2, 3, 4, 5, 6, 7)$, what is the expected consensus value α for each unweighted graph that supports it?
 - (c) Which of the unweighted graphs do you expect to reach consensus the fastest and why? Rank the graphs in the order from fastest to slowest.
3. (15 pts) In this part we will work with continuous-time linear consensus flow

$$\dot{\mathbf{x}} = -L\mathbf{x} \tag{1}$$

and its discrete-time counterpart

$$\mathbf{x}(k+1) = W\mathbf{x}(k) = (I - \Delta t L)\mathbf{x}(k). \tag{2}$$

- (a) Simulate the continuous-time consensus flow (1) over each of the four unweighted graphs in Figure 1 for initial condition $\mathbf{x}(0) = (1, 2, 3, 4, 5, 6, 7)$. Be sure to use a higher-order numerical scheme such as a fourth order Runge-Kutta algorithm, for example using the `scipy.integrate.solve_ivp` function in Python. Discuss how your results relate to your calculations from parts 1 and 2.
 - (b) Simulate (1) for the weighted graph(s) you defined in part 1(d) from the same initial condition. Is the outcome of the process different from the unweighted case?
 - (c) For each of the four unweighted graphs, simulate the discrete-time consensus protocol (2) from the same initial condition. How did you choose the discretization step Δt ? Pick one of the strongly connected graphs and simulate the same discrete-time process with a smaller Δt . Compare how many rounds of communication it took to converge to consensus with the two choices of Δt .
4. (10 pts) In practice, communication networks in robotics often change over time as robots move around. In such a time-varying case, a linear consensus protocol reads

$$\mathbf{x}(k+1) = W(k)\mathbf{x}(k) = (I - \Delta t L(k))\mathbf{x}(k). \tag{3}$$

where the structure and weights in $W(k)$ can change with time. Consider a scenario in which the communication graph can switch between the two graphs pictured in Figure 1(a) and 1(b). Simulate this scenario (you can pick your own switching rule, for example at each k one of the graph is chosen with probability p , or the graphs switch periodically every m steps). Is consensus eventually reached? How does the speed of convergence compare to the static graph case?

Problem 2: Distributed Estimation and Sensor Fusion (50 pts)

A common task in robotics is to construct a map of an unknown environment from noisy sensor measurements. In this problem, we will imagine a group of four robots collaboratively constructing an occupancy map of the room pictured in Figure 2. Black squares indicate that an object is present in the indicated grid cell, and white squares represent unoccupied space. For simplicity, we will assume that our robots are static sensors and do not move around. The map is provided for you in the `room_2x2.csv` file on Canvas.

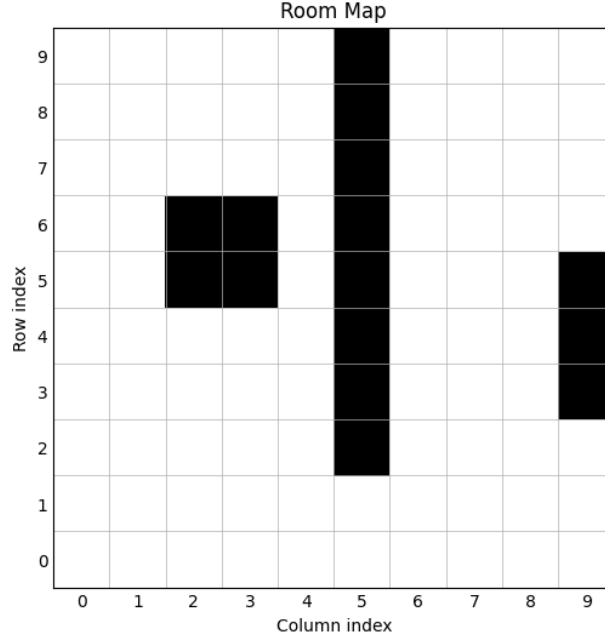


Figure 2: Occupancy map of 10 by 10 space.

Setup. Grid indices are $(r, c) \in \{0, \dots, 9\}^2$. Let $m_{rc} \in \{0, 1\}$ denote occupancy (1 = occupied). Four robots $k \in \{1, 2, 3, 4\}$ are fixed at grid centers

$$\mathcal{P} = \{(1, 0), (1, 8), (7, 0), (8, 8)\}.$$

Robots have unlimited range but only observe cells in line-of-sight (LoS) within the grid (endpoints may be occupied). At discrete times $k = 1, 2, \dots, T$ each robot i produces a binary measurement $z_{rc}^{(i)}(k) \in \{0, 1\}$ for each LoS cell (r, c) it can see, drawn independently across cells and robots from:

$$\Pr(z = 1 \mid m = 1) = p_{\text{hit}}, \quad \Pr(z = 1 \mid m = 0) = p_{\text{fa}}, \quad (4)$$

with hit rate $p_{\text{hit}} = 0.85$ and false alarm rate $p_{\text{fa}} = 0.10$. Cells not in LoS are not measured by that robot.

We assume two robots can exchange messages if they are within line of sight of one another, defining the line of sight graph $G = (V, E)$. We will build the discrete-time averaging consensus matrix W using Metropolis weights where $d_i = |\mathcal{N}_i|$ is the degree of robot i on the communication graph, and

$$w_{ij} = \begin{cases} \frac{1}{1 + \max\{d_i, d_j\}}, & (i, j) \in E, \\ 1 - \sum_{j \in \mathcal{N}_i} w_{ij}, & j = i, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Each robot i associates with each grid cell (r, c) a *belief* $x_{rc}^{(i)}$ which is updated over time to incorporate new sensor measurements and social information. We will be using a binary Bayes filter to update these beliefs. Instead of storing a cell's occupancy as a probability $p_{rc}^{(i)} \in [0, 1]$, we keep its *log-odds*

$$x_{rc}^{(i)} = \log\left(\frac{p_{rc}^{(i)}}{1-p_{rc}^{(i)}}\right), \quad (6)$$

which turns Bayesian updates into simple additions. Starting from a prior p_0 (so $x_{rc}^{(i)}(0) = \log \frac{p_0}{1-p_0}$), each independent measurement $z_{rc}^{(i)}(k)$ adds a fixed “evidence” increment to the log odds:

$$\hat{x}_{rc}^{(i)}(k+1) = x_{rc}^{(i)}(k) + \begin{cases} \log\left(\frac{p_{\text{hit}}}{p_{\text{fa}}}\right), & z_{rc}^{(i)}(k) = 1, \\ \log\left(\frac{1-p_{\text{hit}}}{1-p_{\text{fa}}}\right), & z_{rc}^{(i)}(k) = 0. \end{cases} \quad (7)$$

When needed, we can convert from log odds back to probability with

$$p_{rc}^{(i)} = \frac{1}{1 + e^{-x_{rc}^{(i)}}}. \quad (8)$$

1. **(10 pts)** Implement a Python function that, given two grid indices (r_1, c_1) and (r_2, c_2) and the map, returns **True** if the two cells are connected by a straight line that does not pass through the interior of any occupied cell (excluding the two endpoints), and **False** otherwise. Implement a second python function that, given grid index (r, c) and the map returns a measurement $z_{rc} \in \{0, 1\}$ following the probabilistic measurement model (4). For both of these functions, assume that any cell containing a robot is also occupied.
2. **(10 pts)** For each robot i , compute its *visible set* $\mathcal{V}^{(i)} = \{(r, c) : \text{LoS}((r, c), \mathcal{P}_i) = \text{True}\}$ using the ground-truth map file and the function you wrote in part 1. Visualize the visible set for each robot.
3. **(10 pts) Local Bayesian update (log-odds).** To construct the occupancy map, our robots will first perform local belief updates incorporating new information from their sensor measurements. We will assume that the robots begin with no knowledge of the environment, and initialize each robot's per-cell log-odds with prior $x_{rc}^{(i)}(0) = \log\left(\frac{p_0}{1-p_0}\right)$ using $p_0 = 0.5$.

Write a function that performs one step of a Bayesian belief update for robot i . The inputs to the function should be the array $\mathbf{x}^{(i)}(k)$ containing 100 beliefs of robot i corresponding to the 100 grid cells in the occupancy map (you could choose to store these as a vector or a matrix, depending on your preferences), as well as the visible set $\mathcal{V}^{(i)}$ of robot i . For each measured cell $(r, c) \in \mathcal{V}^{(i)}$, take a measurement of the occupancy using the measurement function you wrote in Part 1. Then update robot i 's belief according to the Bayesian update (7); set $\hat{x}_{rc}^{(i)}(k+1) = x_{rc}^{(i)}(k)$ for unmeasured cells.

4. **(10 pts) Consensus step.** After local updates, robots diffuse their beliefs across the graph using averaging consensus. Write a function perform one step of averaging consensus over the communication graph for *each cell independently*, using the Metropolis weights defined in (5):

$$x_{rc}^{(i)}(k+1) = \sum_{j=1}^4 w_{ij} \hat{x}_{rc}^{(j)}(k+1).$$

Write a simulation that runs the loop “measure $\rightarrow$ local update $\rightarrow$ consensus” for all four robots for $K = 20$ iterations.

5. **(10 pts)** Convert log-odds beliefs of all four robots at the final step K of the simulation to occupancy probabilities using the inverse relationship (8). Plot (i) probability heatmaps and (ii) binary maps for each robot (choose a probability threshold to produce the binary maps, for example 0.6). How would the outcome of the estimation be different if there was no cooperation between robots?